PyGameMaker — Tutorial 1: First Steps — Teacher Guide

Companion for leading students through the in-app tutorial (**Help > Tutorials > First Steps**, 4 pages) and the matching student handout. No programming experience is assumed for you or the students.

Overview

This first tutorial is an interface tour plus one hands-on project. By the end, every student will have a project with one sprite, one object, one room, and will have pressed **F5** to see it run.

> The object students place will not move or react to anything yet — events and behaviors are introduced in Tutorial 2 ("First Game"). If a student asks "why doesn't it do anything?", that is the expected, correct result for this lesson, not a mistake.

Suggested Timing (about 45 minutes)

These are starting points — adjust freely to your class and lab setup.

Segment	Time	What happens
Welcome & discussion	5 min	Open the tutorial panel (page 1), talk through what the series covers
Interface tour	10 min	Page 2 — point out the 3 panels live on your own screen/projector
Hands-on: first project	20-25 min	Page 3 — the 8 steps below, walk the room
Test & celebrate	5 min	Everyone presses F5 and sees their object appear
Wrap-up / next time	5 min	Page 4 — preview what Tutorial 2 adds

Step-by-Step Walkthrough & Common Issues

1. **New Project** (File > New Project, Ctrl+N) — Have students save to a location they will remember (their own folder / the class shared drive). Decide this convention before class starts.
2. **Create a Sprite** (spr_player) — The spr_ prefix is a PyGameMaker naming convention, not a requirement, but establishing it now keeps later projects organized.
3. **Edit the Sprite** — Students draw a simple character in the Sprite Editor. Drawing takes longer than anything else in this lesson, so set a time limit (e.g. "2 minutes, just a quick shape") if the session is tight.
4. **Create an Object** (obj_player) — Nothing visual happens yet; that is expected, not an error.
5. **Assign a Sprite** — **Most common mistake**: forgetting to assign the sprite to the object. If a student's character does not appear later, this is almost always why — check the object's Sprite field first.

6. **Rename the Room** (room0 → room_game) — Every new project already contains a room called room0, so students rename it rather than create one. **Common mistake**: adding a second room instead. The game starts in the *first* room, so F5 then opens an empty black window while the object sits in the other room — rename room0, or place the object in it.
7. **Place the Object** — **Common mistake**: opening the room editor but not actually selecting obj_player in the object list before clicking in the room, so nothing gets placed. There is no "correct" position for this exercise — anywhere in the room is fine.
8. **Test the Game** (F5 or Build > Test Game) — A window opens showing the room and the placed sprite. It will not move or respond to keys yet — that is next lesson.

> The Test Game window. When a student presses F5, the PyGameMaker editor window automatically minimizes itself for the duration of the test and comes back when the student closes the game window. This stops the game window from getting lost behind the editor. On most lab machines the editor will also jump back to the front by itself; on a few Linux desktop setups it may only flash/highlight in the taskbar instead — if that happens, tell students to just click on it once to bring it forward.

Vocabulary Introduced

Term	What it means here
Asset	Any resource in your project: a sprite, sound, object, or room
Sprite	An image (or animation) used to draw something on screen
Object	A game entity — what actually appears and behaves in a room
Room	A game level or screen; where objects are placed

If Time Allows / Sneak Peek

From the tutorial's own "Next Steps" page: next lesson adds **movement** (a Keyboard event) and introduces **Blockly** visual programming. If a student finishes early and asks what is next, it is fine to mention this by name, but there is no need to have them attempt it today — there is not yet an event to attach it to.

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every student pressed F5 and saw their object appear in the room
- [ ] Note which students want to keep exploring — good candidates to pair with a struggling classmate next session

> Full tutorial series: Help > Tutorials inside PyGameMaker. Written documentation and more sample projects: the project wiki.

Worksheet Answer Key

Part A: 1-B, 2-D, 3-C, 4-A.

Part B: 1. The Asset List (left panel). 2. The Sprite Editor (Editor Area, opened by double-clicking the sprite). 3. The Properties panel (right). 4. The toolbar (or **Build > Test Game**, or F5).

Part C:

1. An object is only a description until it is placed in a room; the room is what the game actually shows. Accept any answer that separates "made" from "placed".
2. Any two of: the object has no sprite assigned; the object was never placed in the room; the wrong room was tested; the sprite is empty or transparent.
3. No. Movement needs a Keyboard event with an action, which is introduced in Tutorial 2. Students who expected movement have understood correctly that objects can move; the missing piece is the event.

Parts D and E: completion and reflection; no single correct answer. Use Part E to pick pairs for next session (a student who found something easy with one who found it hard).

Rubric: The First Project

Level	What the project shows
4 - Complete	Saved project with a named sprite, an object using that sprite, a room with the object placed, and a game that runs with F5; names follow the spr_ / obj_ / room_ pattern
3 - Working	Runs with F5 and the object is visible; one naming or organisation slip
2 - Partly there	All assets exist, but the object is missing from the room, or has no sprite
1 - Started	Project created, but fewer than three of sprite, object, room exist